Creative Computing Benchmark

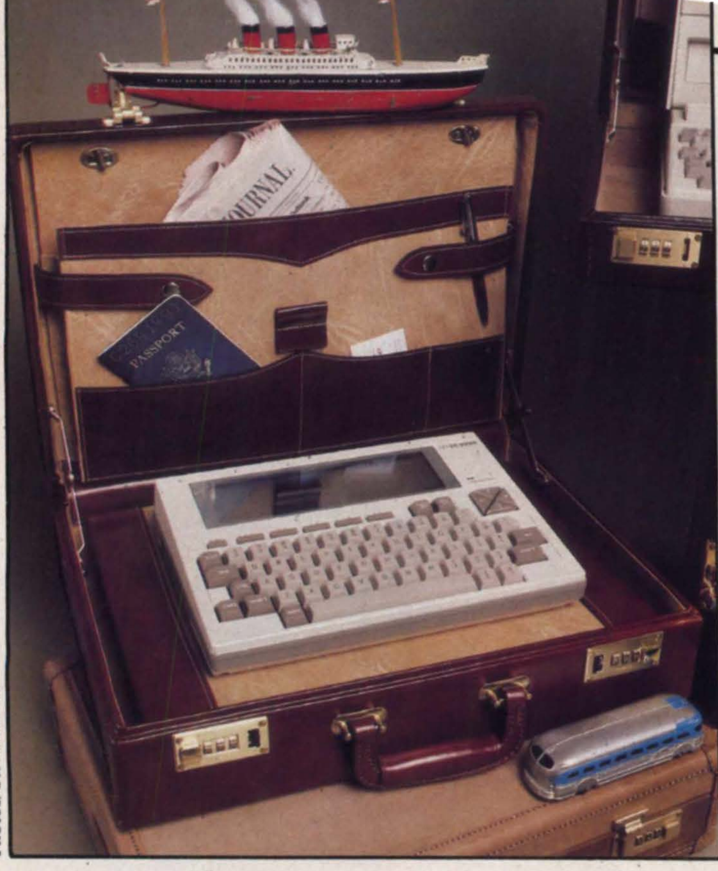
The Creative Computing benchmark is a short test of computational speed, accuracy, and the random number generator in Basic. Computers in the chart are listed in ascending order of completion time of the test expressed in minutes and seconds. In the accuracy measure, the smaller the number the better (.0000001 is excellent while .187805 is poor). In the randomness measure, smaller is better (numbers under 15 are good and over 15 are fair).

Since running the short article about the benchmark test, we have been overwhelmed with responses from readers who ran the test on machines not listed in our original table. With letters still pouring in, here are the results for 107 different computers.

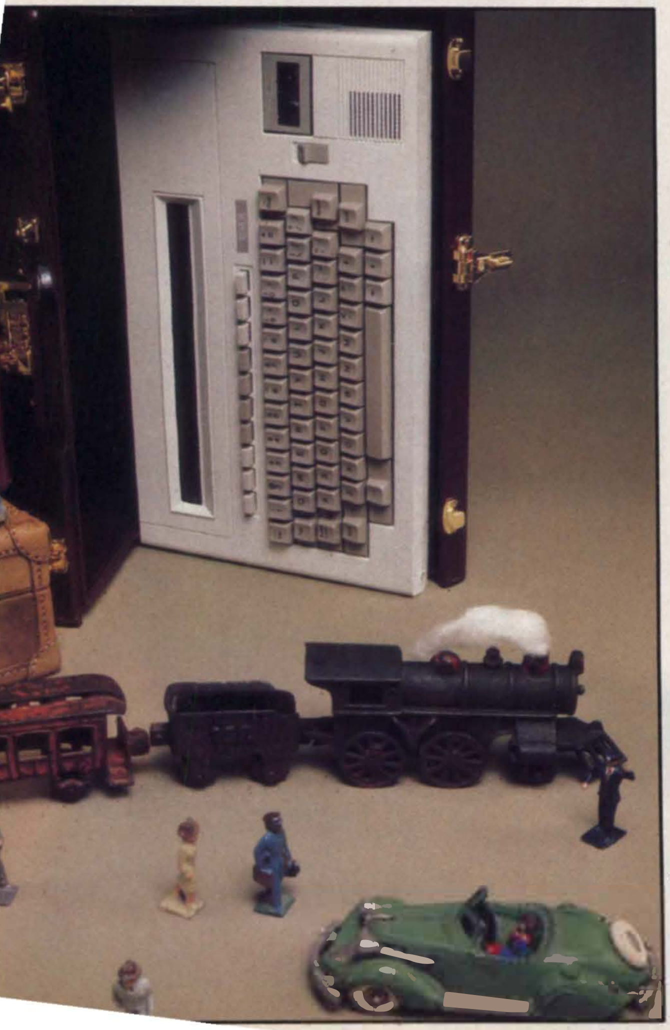
We have taken note of the criticisms of this simple test and are in the process of devising a more comprehensive one. Watch for a follow-up article.—DHA

```
10 ' Ahl's Simple Benchmark
20 FOR N=1 TO 100: A=N
30 FOR I=1 TO 10
40 A=SQR(A): R=R+RND(1)
50 NEXT I
60 FOR I=1 TO 10
70 A=A^2: R=R+RND(1)
80 NEXT I
90 S=S+A: NEXT N
100 PRINT ABS(1010-S/5)
110 PRINT ABS(1000-R)
```

Computer	Time	Accuracy	Random
DEC VAX 11/780	0:01	.0113525	5.3
DEC VAX 11/780 (double)	0:015	.000000000163283	5.3
HP 9845B (390 bit slice)	0:03	.00000882	23.1
Control Data Cyber 730	0:03	.00000000355	6.1
Amdahl 470	0:04	.00000000011846	12.4
HP 3000 Series 44	0:04	.112549	12.9
HP 9836	0:05	.000000000127329	5.5
Wang 2200 SVP	0:05	.000000076	3.9
Stearns Micro	0:08	.005859375	7.1
Burroughs B20	0:09	.005938744544977	3.2
Alpha Micro AM 100T	0:10	.00000387337	12.4
Burroughs B22	0:12	.005859375	15.7
NEC Adv Pers Comp	0:12	.005859375	7.2
Tektronix 4054	0:12	.000000014042598	8.5
Olivetti M20	0:13	.0114136	6.2
Saybrook 68000 (in Apple)	0:13	.00000000011	10.4
TI Professional	0:15	.005859375	7.1
Compaq	0:15	.005859375	7.1
HP 9845B	0:15	.00000882	23.1
Zenith Z-100 (8088)	0:17	.005859375	9.7
ACT Apricot	0:18	.005859375	7.2
Sharp PC-5000	0:18	.005859375	7.2
Eagle PC-2	0:19	.005859375	7.2
DEC Rainbow 100	0:20	.005859375	7.2
Acorn BBC Computer	0:21	.000128746033	5.2
Columbia MPC	0:21	.005859375	7.2
Computer Devices DOT	0:22	.005859375	7.1
IBM PC	0:24	.01159668	6.3
GCE Vectrex	0:33	.0753174	0.9
TI DS990/12 (Mini TS)	0:36	.000000388	3.1
Laser 2001	0:40	.0003272295	17.4
Memotech MX-512	0:46	.00025296211 2	6.9
HP 9020C	0:48	.000000000127329	23.2
Lobo Max-80	0:48	.0338745	5.8
Lynx	0:51	.155	14.1
TRS-80 Model 4	0:53	.0670776	6.5
Panasonic JR200	0:57	.00021481514	15.1
SCS 100	0:59	.187805	7.4
IMS 8000	0:59	.187805	9.6
Alspa ACI-1	0:59	.187805	7.4
DECmate II	0:59	.187805	7.4
Xerox 820-II	0:59	.187805	7.4
Vector Graphic 3 VIP	1:04	.0338745	7.5
Zenith Z-100 (8085)	1:04	.187805	9.5
Toshiba T100	1:09	.187805	7.4
Epson QX-10	1:09	.187805	7.4
Osborne 01	1:10	.187805	7.4
Mattel Aquarius	1:17	.187805	10.0
Epson QX-10	1:18	.187805	7.4
HP-85A	1:20	.00000002	5.8
Morrow MD3 (Bazic 10)	1:21	.000473	3.6
HP-86A	1:25	.00000002	5.8
Tektronix 4051	1:26	.000000014042598	8.1
Digital Group Bytemaster	1:27	.000002779	3.6
NEC PC-8001A	1:29	.0338745	3.0
Atari 800 (MBasic)	1:35	.150879	2.1
Kaypro II	1:36	.187805	7.5
Sony SMC-70	1:37	.0000000458	3.8
HP-75C	1:38	.00000002	5.8
North Star Horizon(10 dig)	1:41	.000473	3.6
NEC PC-8201	1:44	.187805	9.3
MicroOffice RoadRunner	1:48	.187805	7.4
Teleram 3000	1:48	.187805	7.4
Apple III	1:48	.011914	6.7
Vic 20	1:49	.0010414235	23.7
HP 98308	1:52	.00000889	13.1
Commodore 64	1:53	.0010414235	8.9
Apple II plus	1:53	.0010414235	12.0
Apple IIe	1:53	.0010414235	12.0
NEC PC-8801A	1:54	.187805	7.4
Rockwell Aim 65	1:56	.00104141235	14.7
Compucolor II	1:57	.0338745	1.4
TRS-80 Model III	1:59	.0338745	5.8
Micro Color Computer	1:59	.000596284867	7.6
Commodore CBM 8032, 2001	2:01	.0010414235	1.4
Heath/Zenith H-89A	2:04	.187805	7.4
Atari 2600 Graduate	2:15	.0002246797 08	7.9
TRS-80 Model I	2:19	.0338745	12.0
Color Computer	2:23	.000596284867	7.3
Atari 800 (fastchip)	2:23	.006875	7.0
Dragon 32	2:29	.000596284867	7.3
Epson HX-20	2:36	.0338745	23.8
Timex/Sinclair 1000 (fast)	2:43	.00041294098	8.7
Interact Model R	2:50	.0338745	8.1
Wang 2210	2:52	.000011432	12.5
OSI Challenger 1	3:07	.0010414235	13.9
SpectraVideo 318/328	3:40	.0000002058	0.7
TI 99/4A	3:46	.00000011	2.6
Radio Shack PC-3	4:00	.00000627	10.9
TI 99/4A, Extended	4:10	.00000011	10.7
Oric-1	4:10	.00104141235	12.1
Datapoint 1800	4:16	.0000012042	11.3
Sinclair ZX81	4:23	.0006685257	6.3
Sinclair Spectrum	4:39	.0006685257	3.5
TRS-80 Model 100	4:54	.0000002058	0.7
Casio FP-200	5:05	.00723	30.3
Sharp PC-1500 (RS PC-2)	5:10	.0000288	7.8
TI CC-40	5:41	.00000011	6.2
Sanyo PHC-25	5:41	.000267505646	10.2
Canon X-07	6:03	.0000002058	24.9
Atari 1200XL	6:45	.013959	5.2
Atari 400/800	6:48	.012959	22.8
Sharp PC-1250	11:14	.0000288	5.9
Timex/Sinclair 1000(slow)	16:55	.00041294098	7.4
IBM System 23	19:00	.00000005503	3.4
HP-97	23:00	.000034	---
Sharp PC-1211	28:32	.00002882	---



Photos/Steve Borns



CHOOSING A

David H. Ahl

NOTEBOOK COMPUTER

Eight years ago, several of us paid a visit to Alan Kay at the Xerox Palo Alto Research Center (PARC). Late in the afternoon, the mood was given over to blue-sky dreaming, kind of a "wouldn't it be nice if. . ." session. One dream that seemed to be shared by everyone in the group, managers, researchers, and educators alike was that it would be wonderful to have a truly portable computer about the size of a three-ring binder with computational, word processing, and color graphics capability at an affordable price.

Today, except for the color graphics capability, such a machine is here. And it probably won't be too long until the graphics are available too.

Such a computer is not necessarily for everyone, but it is far easier to name people who could make good use of such a machine than those who could not. Anyone who works with numbers knows the benefits of a computer or programmable calculator. Most writers are switching to word processing systems in the home office. Now even the field notebook and pocket tape recorder can be replaced.

But notebook computers will find application with scores of people who have never touched a computer before—sales people making presentations to clients, business people at a branch office, students in the library, scorers at athletic games, and just plain folks who need to jot down a note or do a quick calculation.

What Is A Notebook Computer?

In a sense, there are three, or possibly four, categories of portable computers. There is the group that first took the name portable—the Osborne, Kaypro, Compaq type of machine. We prefer to think of these sewing-machine size computers as "transportable" rather than truly portable. Most weigh well over 20 pounds and are not something you would want resting on your lap for an extended period of time. Their appeal for most users is something other than portability. These machines are not included in our roundup.

At the other end of the spectrum are the pocket computers such as the Sharp PC-1500, Casio FX-700P, and Radio Shack PC-1. These are capable little units for computational applications, but rather limited for general purpose computing. We have included two machines in this roundup that overlap the pocket category, the TI CC-40 and HP-75C, but we have excluded the true pocket units.

Between these two extremes lie the notebook computers. The forecasters at Future Computing see this group as being split between the Radio Shack Model 100 type of computer and machines with more extensive capabilities such as the Sharp PC-5000, Gavilan, and Grid Compass. We have chosen to ignore this split, although with the rapid migration to this part of the market, perhaps we, too, will be seeking new categories before long.

At the time we put this article on our editorial schedule, there were six notebook computers. In the next three months, ten additional machines were introduced, with two introduced the week before our deadline. Of this total of 16 machines, 14 are included in this roundup. The two that are not included are the Grid Compass and Universal Data UDI-500. We excluded the Compass because its $9000 price was more than twice that of the next machine down, and we felt that it had appeal for a rather different market. We excluded the UDI-500 because we were unable to get the data on it in time for our print deadline.

The Same and Different

The 14 notebook computers are similar in some ways, but quite different in many other ways. All are portable, although 1½ pounds is a great deal more portable than 11 pounds. All perform computations, although the fastest is a staggering 20 times faster than the slowest. The minimum amount of user programmable memory ranges from 8K to 128K, a 16 to 1 difference.

Most use a proprietary operating system, although in most cases it is not an operating system at all, but simply a traffic cop to direct information flow. All but one of the machines speak Basic, mostly Microsoft; many have communications capabilities; but fewer have word processing or spreadsheet packages available.

From these few comparisons, you can see that it is not just a matter of deciding you want a notebook computer, marching down to your local computer store, plunking down your money, and taking one home. As with any other computer purchase, you should have some idea of what you want to do with it, and then look for a machine to meet both your needs and your budget. Also, as with other computers, you almost certainly will find more uses for the machine than you originally anticipated; thus you should buy as much capability as you can comfortably afford.

As you get acquainted with notebook computers, you will find that the manufacturers have made all sorts of tradeoffs. Size versus extra features is an obvious tradeoff—you just can't fit a large display, modem, and printer in a package the size of a paperback novel. Price is a tradeoff against nearly everything—speed, memory capacity, and technical sophistication. In the following sections, we discuss some of the key features—and related tradeoffs—in more detail.

The Display

Every one of the 14 notebook computers uses a liquid crystal display (LCD), and three of them (Gavilan, Sunrise, and Toshiba) have the capability to drive a CRT monitor as well. Note that these are also three of the highest priced units.

The LCD display is the familiar black on gray found in most digital watches and calculators today. A key advantage to an LCD display is that it draws relatively little power and is more or less shock resistant. It is light in weight compared to a CRT, but, because of the rigid material required for the display, larger ones start to add pounds, not ounces, to the weight of a computer.

LCD displays have several disadvantages. First, the response time is much slower than a CRT. Second, the pixels are relatively large thus ruling out high-resolution graphics. Third, in certain lighting conditions, LCD displays are difficult to read, even though most of the computers provide both tilt and contrast adjustments. And fourth, LCD displays are very sensitive to cold; at temperatures below freezing, they don't work reliably, and may stop altogether.

The Gavilan, RoadRunner, and Sharp display the most text characters, although the physical size of the displays on these machines is not the largest. This means that the characters are quite small—about the same size as dot matrix printer output. The WorkSlate, a spreadsheet-only machine, displays even more characters in its higher 16-line x 46-character display; it is more readable.

The displays on the remaining machines use larger pixels; they thus have larger character sizes and are more legible. On the other hand, computers with

LCD Display

Computer	Lines x	Chars.	Total	Graphics Pixels
WorkSlate	16	46	736	none
Sharp PC-5000	8	80	640	80x640
RoadRunner	8	80	640	64x480
Gavilan	8	80	640	64x400
Toshiba T100	8	40	320	64x240
NEC PC-8201	8	40	320	64x240
Radio Shack 100	8	40	320	64x240
Teleram 3000	4	80	320	none
Xerox 1810	3	80	240	32x480
Casio FP-200	8	20	160	64x160
Epson HX-20	4	20	80	32x120
Canon X-07	4	20	80	32x120
HP-75C	1	32	32	none
TI CC-40	1	31	31	none

Time and Accuracy

Computer	Time	Accuracy
Sharp PC-5000	0:18	.005859375
Gavilan*	0:20	.005859375
Toshiba T100	1:09	.187805
HP-75C	1:38	.00000002
NEC PC-8201	1:44	.187805
RoadRunner	1:48	.187805
Teleram 3000	1:48	.187805
Xerox 1810	1:48	.187805
Epson HX-20	2:36	.0338745
Radio Shack 100	4:54	.0000002058
Casio FP-200	5:05	.00723
TI CC-40	5:41	.00000011
Canon X-07	6:03	.0000002058
WorkSlate**	n/a	n/a

* Estimated values
** Runs only spreadsheet software

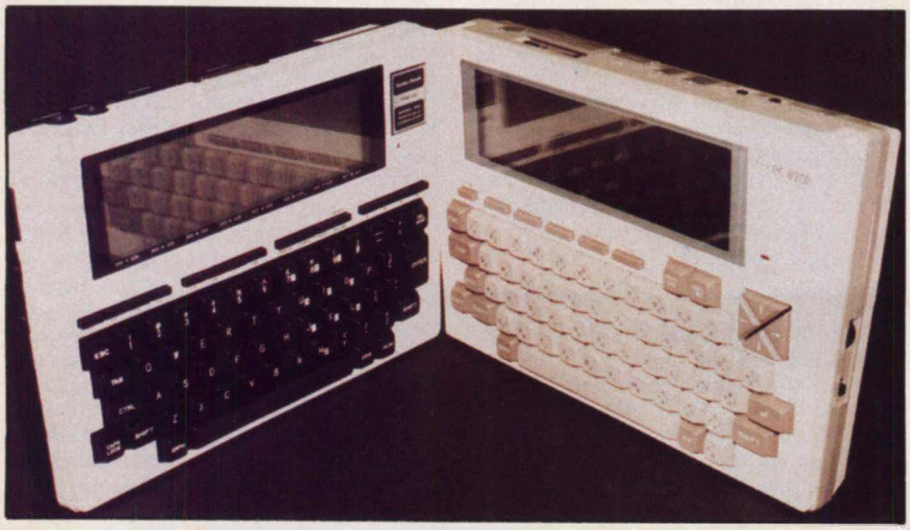
one-line displays (HP and TI) are not usable for word processing, and are barely usable for spreadsheet work or even Basic programming. Even the two machines with 4-line by 20-character displays (Canon and Epson) are difficult to use and have been eclipsed by the larger LCD displays.

It is unlikely that you would want to use a notebook computer for any serious graphics except simple plots and bar charts. Ten of the computers provide such rudimentary plotting capability (see chart); we judge the Canon and Epson unsatisfactory for graphs even though Basic allows pixel addressing.

Four representative notebook computers: Teleram T3000, Radio Shack 100, TI CC-40, and Epson HX-20.

Speed and Accuracy

We were unable to run our benchmark on three of the 14 computers; however, based on the mpu, clock rate, and operating system, we were able to estimate the values for the Gavilan and Sunrise. As Basic is not offered on the WorkSlate, it is not included in the chart.



The Radio Shack 100 and NEC PC-8201 are twins in some regards, but they have their differences.

The two 16-bit 8088-based machines, the Sharp and Gavilan, are the speed demons. Both use MS-DOS with Microsoft GW (Gee Whiz) Basic. The Toshiba, NEC, and RoadRunner also use Microsoft Basic, but with an 8-bit 80C85 mpu, the CMOS version of the Z80A. Accuracy is grim, but speed isn't bad.

The Epson also uses Microsoft Basic, but on a different Z80 look-alike, a CMOS 6301 mpu. It is slightly slower, but considerably more accurate than the 80C85.

The Radio Shack Model 100 is a virtual twin of the NEC 8201 (both are made by Kyoto Ceramics in Japan), but Radio Shack elected to use double precision variables as the default mode. Hence, it is 2½ times slower than the NEC, but has astoundingly greater accuracy. The Canon X-07 uses a similar approach but, because of running the mpu at a slower clock rate, is even slower.

Incidentally, double precision variables can be specified on the NEC,

All of the notebook computers fit in an attache case. Here is one of the largest, the Toshiba T100, in a case with LCD display, acoustic modem, memory cartridge, and cables.

Toshiba, and RoadRunner and will yield the same speed and accuracy as the Model 100. The reverse is not true; specifying single precision on the Model 100 results in only about an 11% improvement in speed.

For a small machine, the HP-75 is amazingly fast and accurate; indeed it is the most accurate of all the computers we have ever tested, including several running in double precision mode.

The Casio and TI are the leisurely performers, but TI Basic with its 16-bit TMS 9995 mpu has the second best accuracy of all the computers tested, a nice plus for engineers and scientists who need accurate results to seven decimal digits.

Memory and Mass Storage

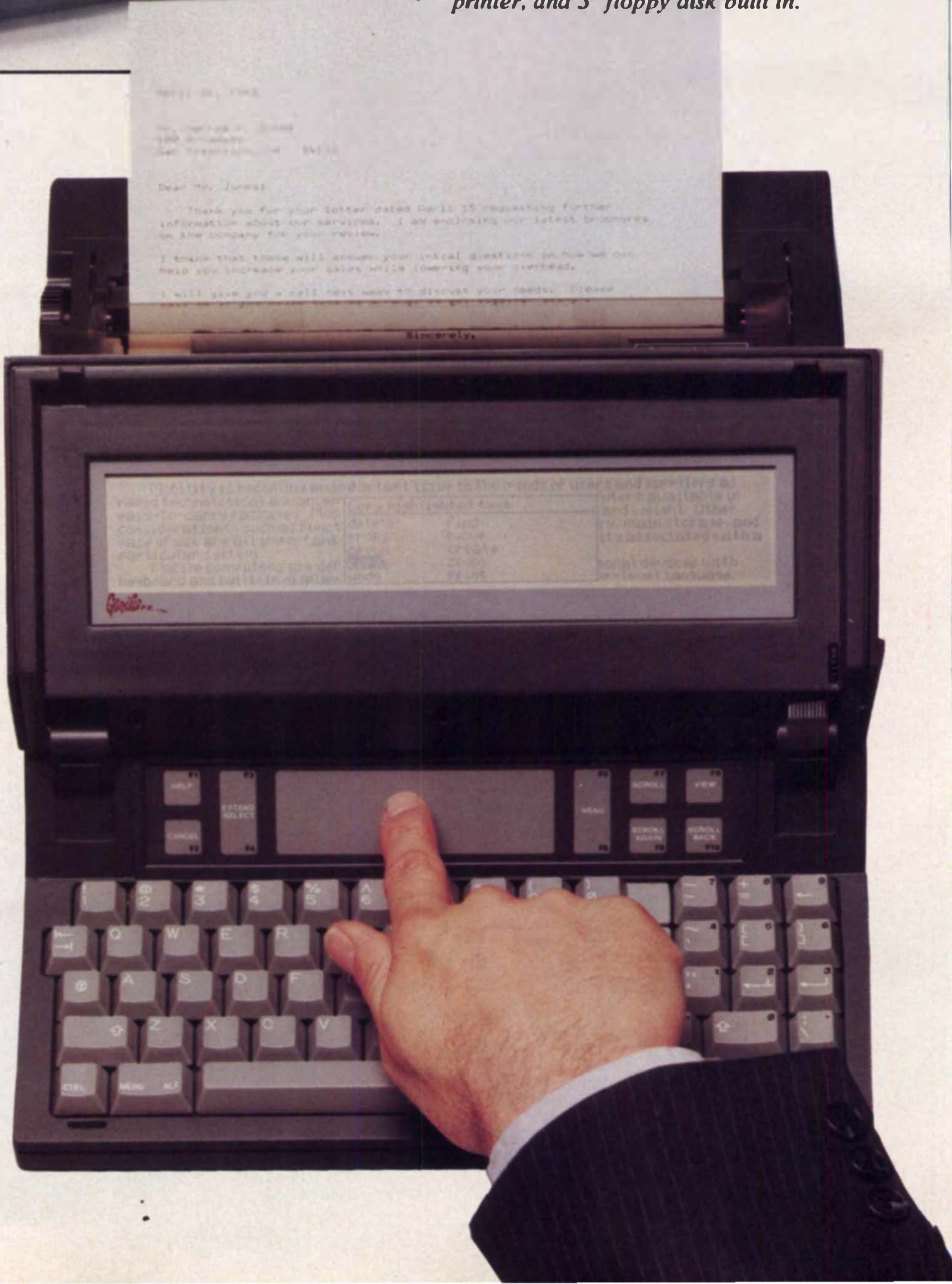
Four of the basic computers come with 8K of memory or less. Our experience indicates that this is barely adequate for any but the simplest applications. Two of the computers, the TI and the WorkSlate, can be expanded only to 16K; this, too, may prove limiting.

The upper internal memory limit on the next five machines is 32K, which should be adequate for most jobs. However, if you intend to leave several large programs or data files permanently or temporarily in the computer, you will probably want even more memory. The two 16-bit computers, the Sharp and Gavilan, provide the most internal memory capacity.

It is in the area of mass storage that manufacturers have taken radically different approaches. The HP-75C uses small magnetic cards, very convenient, but with rather limited capacity. Three of the computers have built-in microcassette recorders, the Epson, Xerox, and WorkSlate. This is a very satisfactory approach with a notebook computer as it does not require an external recorder with its added bulk and messy cables to be dragged around.

Another satisfactory, but more expensive approach, is plug-in RAM cartridges. The cartridges on the NEC, Toshiba, and RoadRunner are standard CMOS memory with a lithium battery. Even more expensive, but higher capacity bubble memory cartridges are available for the Sharp and Teleram.

The full-featured Gavilan has a touch pad, printer, and 3" floppy disk built in.



Unique among the 14 machines is the Gavilan with a built-in 3" floppy disk drive. Obviously, it must be treated with added care since a mechanical drive is inherently more sensitive to shock than a solid state device. Since none have been in the field long enough, we cannot comment on reliability.

Two of the computers, the Sharp and Toshiba, provide an interface to standard floppy disk drives for operation at home base. Several of the others plan to offer optional floppy disk interfaces.

Computer	RAM Memory Min	RAM Memory Max	Mass Storage
Sharp PC-5000	128K	256K	Bubble memory cartridges, external cassette or disk
Gavilan	64K	336K	3" microfloppy
Teleram 3000	64K	64K	Bubble memory cartridges
Toshiba T100	64K	64K	RAM pack, ext. cassette, or floppy disk
RoadRunner	48K	48K	RAM cartridges
Xerox 1810	16K	64K	Microcassette
NEC PC-8201	16K	64K	RAM pack, ext. cassette
Epson HX-20	16K	32K	Microcassette
WorkSlate	16K	16K	Microcassette
Casio FP-200	8K	32K	RAM pack, ext. cassette
Radio Shack 100	8K	32K	External cassette
HP-75C	8K	32K	Magnetic card
Canon X-07	8K	24K	RAM card, ext. cassette
TI CC-40	6K	16K	External wafertape

Software, Internal and External

Eight of the 14 computers have proprietary operating systems which, as we said earlier, are generally not operating systems per se, but rather controllers of information and data flow. The two 8088-based machines, the Sharp and Gavilan, have full MS-DOS operating systems with, seemingly, few compromises for the small size.

For of the 8-bit machines have CP/M or CP/M compatible operating systems. These four computers plus the two with MS-DOS are theoretically able to run more existing software packages than the machines with proprietary systems. But it doesn't always work out that way.

The chart shows the availability of Basic (Microsoft, on all but four machines). The calculator-like machines, TI and HP, each use their own Basic. The TI Basic on the CC-40 is similar to Microsoft, but HP Basic has many differences and idiosyncrasies. Casio Basic is similar to Microsoft, and the WorkSlate, of course, has no Basic at all.

The other software packages listed in the chart are those that are built into the basic hardware or are currently available on the appropriate media. On the machines with standard operating systems, other software packages can often be loaded through the RS-232 port and will run with minor modifications. For example, we loaded *WordStar* into our Teleram and got it running with relatively few problems. On the other hand, the menus and CRT screen overlays do not fit the LCD displays on these notebook portables, so even if a package can be loaded, it might not work.

Some of the manufacturers are encouraging the development of software by third party vendors, some are "not discouraging development," and still others plan to do everything internally or under contract and are not encouraging development at all.

Beyond the packages listed in the chart, currently, the most additional software is available for the TI, HP, and Toshiba computers. A goodly amount is available for the Radio Shack 100, and slightly less for the Epson and NEC 8201.

Computer	Operating System	Basic	Spread Sheet	Word Processor	Comm.
Canon X-07	Prop.	MS	-	-	-
Casio FP-200	Prop.	Y	Y	-	-
Epson HX-20	Prop.	MS	-	Y	-
Gavilan	MS-DOS	MS	Y	Y	Y
HP-75C	Prop.	HP	Y	Y	-
NEC PC-8201	Prop.	MS	-	Y	Y
Radio Shack 100	Prop.	MS	-	Y	Y
RoadRunner	CP/M	MS	Y	Y	Y
Sharp PC-5000	MS-DOS	MS	Y	Y	Y
Xerox 1810	CP/M	MS	Y	Y	Y
Teleram 3000	CP/M	MS	-	Y	Y
TI CC-40	Prop.	TI	-	-	-
Toshiba T100	CP/M	MS	Y	Y	Y
WorkSlate	Prop.	-	Y	-	Y

Just How Portable is Portable?

Although notebook computers are considerably more portable than the machines that created the portable label just $1\frac{1}{2}$ years ago, some are much more portable than others. In general, size is inversely proportional to capability and features, but this is not universally true.

Three of the machines weigh under two pounds and are about the size of a hardback novel. Two, the TI and HP, have one-line displays, but one, the Canon X-07, packs a four-line display in this size package.

The WorkSlate is slightly larger but, because of its flat, less-than-full-stroke keyboard, it is just an inch thick. Its 16-line x 46-character display is a nice treat on a machine this small.

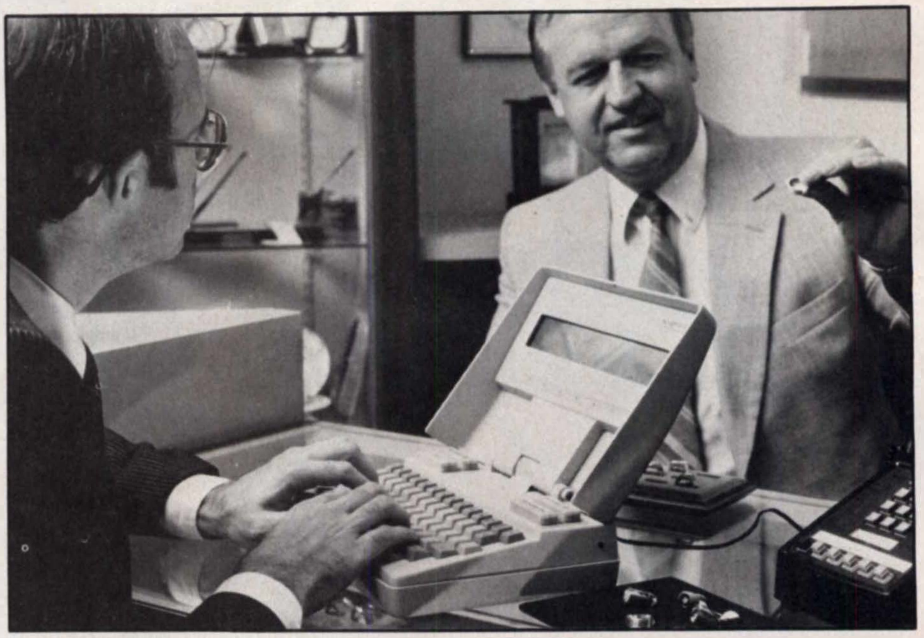
Going up in size, the next six machines (Epson, Radio Shack, NEC, Casio, Xerox, and RoadRunner) all weigh between four and five pounds, and all are the size of a thick (2" to 3")

Computer	Weight	Size			Volume
Canon X-07	1.1 lb.	8.0" x	5.1" x	1.0"	41 cu. in.
TI CC-40	1.4	9.5	5.8	1.0	55
HP-75C	1.6	10.0	5.0	1.3	65
WorkSlate	3.2	11.3	8.5	1.0	96
Epson HX-20	3.8	11.3	8.5	1.8	173
Radio Shack 100	3.8	11.8	8.5	2.0	201
NEC PC-8201	3.8	11.8	8.5	2.4	241
Casio FP-200	4.0	12.5	8.8	2.2	242
Xerox 1810	5.0	16.0	9.0	2.0	288
RoadRunner	5.0	11.5	7.8	3.0	269
Gavilan	9.0	11.4	11.4	2.8	364
Teleram 3000	8.8	13.0	9.7	3.5	441
Sharp PC-5000	11.0	12.8	12.0	3.4	522
Toshiba T100	13.0	16.5	11.0	4.0	726

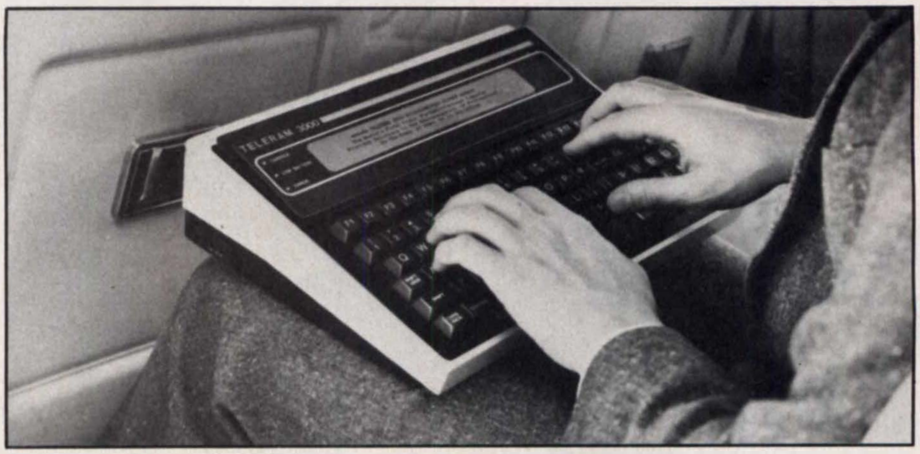
three-ring binder. All will fit easily in a standard attaché case, and weigh no more than an equivalent amount of paper.

The next four machines represent a big jump in size and weight. While none of them approach the size and weight of the sewing machine portables, you will probably think twice before tossing one of them in your briefcase. Ten pounds sounds light, but on long trips your arm will start to feel as though it is stretching.

In some sense, the Toshiba is not a true portable, since it requires an AC power source. The design philosophy was to produce a machine, the cpu of which could be carried around along with a small LCD display, modem, and memory cartridges for work away from the home office. Back home, the same T100 serves as a desktop computer with a stationary monitor and floppy disk drives.



Notebook computers are an excellent sales tool. Here is a RoadRunner in the field.



Notebook computers are excellent travelling companions. Unfortunately, four airlines have banned their use in recent weeks.

Overall Ratings

The overall ratings of any products are, of course, affected by the biases of the testers. This is no exception. We are computer enthusiasts, so we look for good performance. We are writers, so we look for excellent word processing. We travel a great deal, so we look for light-weight portability. We are underpaid, so we look for an attractive price. We are impatient, so we look for fast storage.

In an effort to remove our biases, we rated each computer in each of 12 areas, as objectively as possible. We added up the total points and plotted the resultant value against the price.

The following are the 12 areas of ranking, and how each was scored.

• *Speed:* 7 minus benchmark completion time to nearest minute.

Computer	Speed	Weight	Keyboard	Display	Input/Output	Memory	Mass Storage	Programming	Graphics	Software	Battery life	Features	Total
TI CC-40	1	7	1	0	1	0	2	2	0	5	8	0	27
Casio FP-200	2	6	3	2	2	1	2	4	4	2	3	0	31
WorkSlate	3	6	3	6	2	1	3	1	0	2	3	1	31
Canon X-07	1	7	2	1	3	1	2	3	3	1	8	1	33
HP-75C	5	7	1	0	2	1	2	2	0	5	8	0	33
Teleram 3000	5	4	5	3	2	4	5	4	0	3	2	1	38
Toshiba T100	6	0	5	3	4	4	4	4	5	6	0	1	40
Radio Shack 100	2	6	5	3	5	1	2	3	5	5	2	1	40
Epson HX-20	4	6	5	1	3	1	3	3	3	2	8	2	41
NEC PC-8201	5	6	5	3	5	1	4	4	5	3	2	0	43
RoadRunner	5	5	5	5	3	3	4	3	4	4	2	1	44
Xerox 1810	5	6	5	2	4	2	3	4	5	4	2	3	45
Sharp PC-5000	7	3	5	5	4	8	5	5	5	4	2	1	54
Gavilan	7	4	5	5	4	5	5	5	5	4	2	5	56
A ______													
B ______													

• *Size/Weight:* 8 minus weight divided by 2, rounded off.

• *Keyboard:* 5, standard full-stroke; 4, non-standard; 3, full-size, not full-stroke; 2, smaller; 1, calculator style.

• *Display:* Characters in display divided by 120, rounded off.

• *I/O:* Number of I/O ports.

• *Memory:* Minimum internal memory divided by 16K.

• *Mass Storage:* 5, floppy or bubble memory; 4, RAM cartridge; 3, built-in cassette; 2, external cassette; 1, other.

• *Programming Ease:* 5, large display, on-screen editing; 4, medium display, on-screen editing; 3, small display, on-screen editing or no on-screen editing; 2, one-line display; 1, no programming.

• *Graphics:* display height in inches plus number of vertical pixels divided by 20, rounded off.

• *Software:* Packages currently available; maximum 6.

• *Battery Life:* Life in hours divided by 5, maximum 8.

• *Other Features:* Count of number of additional features (such as built-in floppy disk, built-in printer, etc.)

Our Chart, And Yours

At the bottom of the ratings chart, we left room for two additional computers. Here you can rate the new ones that inevitably were introduced the day this magazine went to press. Or you can rework some of our figures.

After computing the ratings, we plotted the overall total against the suggested list price. Five of the machines are practically sitting on the price/performance curve. They are the Casio, WorkSlate, HP-75C, RoadRunner, and Xerox. All are priced fairly for their capability and features.

The computer that was introduced the earliest in this group of 14 is the Teleram 3000. At the time of its introduction, it represented a breakthrough on many fronts, and was priced accordingly. Today, however, a different competitive climate exists, and the Teleram, at $2495, appears to be overpriced for what it delivers.

The Toshiba T100 is also over the curve, but for a different reason. As we

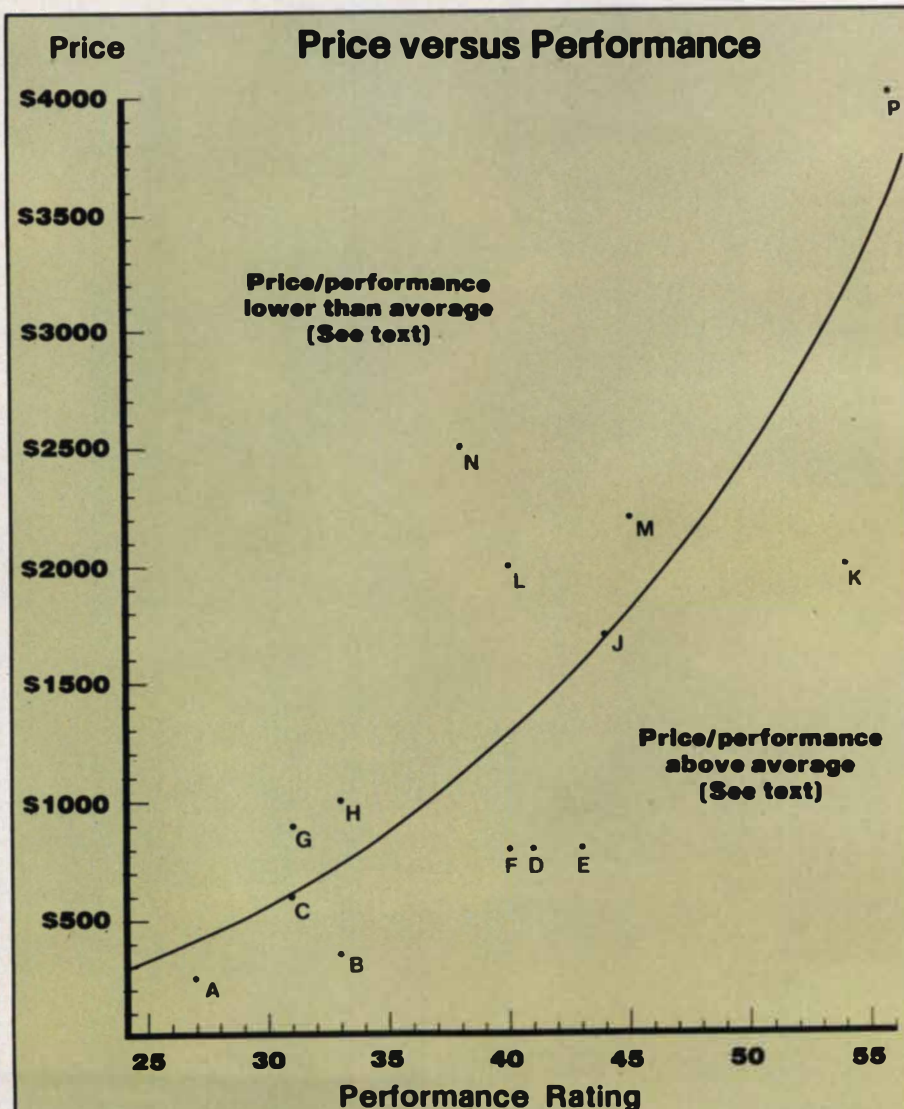


Table 1. Chart plots price (vertical) against performance (horizontal). Curve shows average for 14 notebook computers.

A	TI CC-40	F	Radio Shack 100	L	Toshiba T100
B	Canon X-07	G	WorkSlate	M	Xerox 1810
C	Casio FP-200	H	HP-75C	N	Teleram T3000
D	Epson HX-20	J	RoadRunner	P	Gavilan
E	NEC PC-8201	K	Sharp PC-5000		

mentioned earlier, the T100 is the cpu component of a full-fledged desktop machine, one which we rated very highly in our review in November. Hence, it should not be compared only to portables, but should be viewed in its dual role. In this light, it may well represent a bargain.

Also above the curve is the Gavilan. This is a state-of-the-art machine that offers many novel and unique features—built-in 3" floppy disk, touch pad, snap-on printer, and Lisa-like software. Obviously, this all costs money, and the Gavilan is priced accordingly.

But perhaps most interesting are the five machines that fall below the curve, and thus represent a relative bargain. At the low end is the TI CC-40. For professionals, students, and engineers, this is an unbeatable machine at only $250, frequently discounted to well under $200.

Although we haven't had an opportunity to give the Canon X-07 a thorough shakedown, at $350 it appears to be a good buy.

Computer	Price
TI CC-40	$ 250
Canon X-07	350
Casio FP-200	499
Epson HX-20	795
NEC PC-8201	795
Radio Shack 100	799
WorkSlate	895
HP-75C	995
RoadRunner	1695
Sharp PC-5000	1995
Toshiba T100	1995
Xerox 1810	2195
Teleram 3000	2495
Gavilan	3995

Coming up a bit, if you can put up with the small screen size of the Epson HX-20, it offers a large amount of capability and extra features such as a built-in microcassette and printer and long battery life.

The Radio Shack 100 and NEC 8201 twins also fall below the curve. In our ratings, the NEC is the preferred machine, mainly because of the plug-in memory cartridges and on-screen Basic editing. On the other hand, if you want a built-in modem, the Model 100 is the machine of choice.

At the high end, the Sharp PC-5000 falls way below the curve. This is a spectacular, state-of-the-art computer offering tremendous capability in the $2000-3000 price range. The only real disadvantage of the Sharp is the 11-pound weight and somewhat greater bulk than many other units. To us, this seems a small drawback against its outstanding performance.

Obviously, our choice is not necessarily yours. Choosing a notebook computer—or any computer for that matter—is not easy. The increasing number of entries in the market and the bewildering array of features make the choice a tough one. Armed with the information from this issue, plus a healthy dose of skepticism and patience, you should be able to find the computer that best meets your needs and budget. ■

Computer,
Issue Reviewed
Manufacturer Name
& Address

Canon X-07
Canon USA
One Canon Plaza
Lake Success, NY 11042
(516) 488-6700

CIRCLE 401 ON READER SERVICE CARD

Casio FP-200
Casio, Inc.
15 Gardner Rd.
Fairfield, NJ 07006
(201) 575-7400

CIRCLE 402 ON READER SERVICE CARD

Epson HX-20
(March 1983, Buyer's Guide 1984)
Epson America, Inc.
3415 Kashiwa St.
Torrance, CA 90505
(213) 539-9140

CIRCLE 403 ON READER SERVICE CARD

Gavilan
Gavilan Computer Corp.
240 Hacienda Ave.
Campbell, CA 95008
(408) 379-8000

CIRCLE 404 ON READER SERVICE CARD

HP-75C
(Buyer's Guide 1984)
Hewlett Packard
1000 N.E. Circle Dr.
Corvallis, OR 97330
(503) 757-2000

CIRCLE 405 ON READER SERVICE CARD

NEC PC-8201
(August 1983, Buyer's Guide 1984)
NEC Home Electronics
1401 W. Estes Ave.
Elk Grove Village, IL 60007
(312) 228-5900

CIRCLE 406 ON READER SERVICE CARD

Radio Shack 100
(August 1983, Buyer's Guide 1984)
Tandy Corp.
Fort Worth, TX 76102
(817) 390-3011

CIRCLE 407 ON READER SERVICE CARD

RoadRunner
(January 1984)
MicroOffice Systems Technology
35 Kings Highway East
Fairfield, CT 06430
(203) 367-2525

CIRCLE 408 ON READER SERVICE CARD

Sharp PC-5000
(January 1984)
Sharp Electronics Corp.
10 Sharp Plaza
Paramus, NJ 07652
(201) 265-5600

CIRCLE 409 ON READER SERVICE CARD

Teleram 3000
(January 1984)
Teleram Communications Corp.
2 Corporate Park Dr.
White Plains, NY 10604
(914) 694-9270

CIRCLE 410 ON READER SERVICE CARD

TI CC-40
(August 1983, Buyer's Guide 1984)
Texas Instruments
P.O. Box 225012
Dallas, TX 75265
(214) 995-3741

CIRCLE 411 ON READER SERVICE CARD

Toshiba T100
(November 1983)
Toshiba America, Inc.
2441 Michelle Dr.
Tustin, CA 92680
(714) 730-5000

CIRCLE 412 ON READER SERVICE CARD

WorkSlate
Convergent Technologies, Inc.
2441 Mission College Blvd.
Santa Clara, CA 95050
(408) 727-8830

CIRCLE 413 ON READER SERVICE CARD

Xerox 1810
Xerox Corp.
Xerox Square 006
Rochester, NY 14644
(716) 423-3539

CIRCLE 414 ON READER SERVICE CARD

When contacting manufacturers, please mention *Creative Computing.*

The Ultimate Notebook Computer

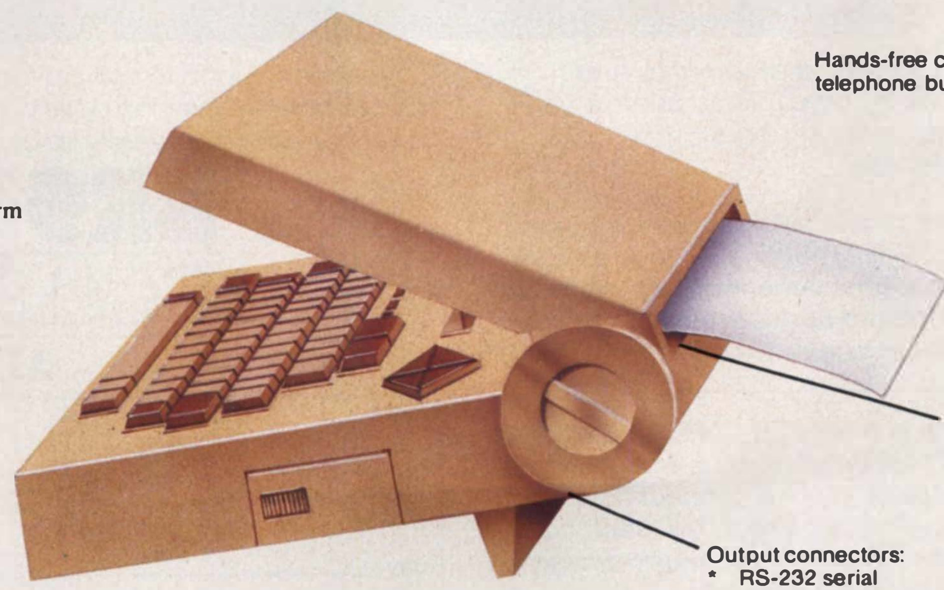
David H. Ahl

While doing the series of articles on notebook computers, speaking to manufacturers, and, most of all, using a notebook computer for the past nine months, I thought of many things that I would like to see on the ultimate notebook computer.

That is not to say that today's machines aren't good—they are. In fact, I firmly believe that notebook computers will have more impact on the way people use computers than anything else since the first computer was invented. They will have more impact than transistors, than minicomputers, than microcomputers, or than graphics displays. Why? Because now you can take a computer anywhere. Your mind extending tool can be your constant companion—and it will be.

But what would be nice to have on it compared to the machines of today? Frankly, just a few additional touches would make today's notebook computers nearly ideal. A little less weight, a touch-sensitive color screen, the ability to run a wide range of software, stereo sound, and a longer battery life would make a mighty fine machine.

We put our heads together, came up with a wish list, and then had John Jinks do a rendering of the Ultimate Notebook Computer. ■




Stereo sound; 3 voices per channel, 5 octaves
Touch-sensitive, flat screen color display (5.5" x 9"). Up to 32 lines of 132 characters; graphics resolution 768 x 1024.
CREATIVE COMPUTING
Ad Size
Inquiries
Index
2-page 9200 213
1-page 5356 124
2/3 4061 94
1/2 3931 91
1/3 2851 66
1/4 2376 55
1/6
1/9
Effect of Color
Color
Index
B&W 100
2-color 120
4-color 138
Requested more info
Purchased product
No action
Plan to buy
Action taken after reading ad
2 Slots for 1 Mbyte non-volatile bubble memory cartridges
Slot for emulation cartridges (Apple, Atari, IBM PC, MSX)
5 Programmable function keys with 4 meanings each
63 Full-stroke alphanumeric and symbol keys in standard layout
Cursor control keys in logical pattern
Toggle for Qwerty and Dvorak keyboards, also numeric keypad
Dual processors (8 and 16 bit)
1 megabyte of internal memory
Portable joystick and mini-mouse (fit in case)
Total compatibility with MS-DOS and CP/M
Contained in ROM:
* Telecommunications
* Word processing
* Spreadsheet
* Database package
* MBasic and CBasic
Optional peripherals:
* 20 Mbyte hard disk
* 5 1/4" and 8" floppy disks which can read and write any disk format (Apple, IBM, etc.)
* Read/write videodisc

14 Notebook Computers in Brief

David H. Ahl

Since the introduction of the first pocket computers just over two years ago and the first transportables about 1½ years ago, we have been avidly following the developments in the field. We have reviewed a cross section of these computers on the pages of *Creative Computing* (the dates of these reviews are noted in the chart at the end of the previous section).

This section is not intended to be an exhaustive, in-depth review of the computers. Rather, it contains a description of each machine including impressions from our review of it or from the comments of other users.

Texas Instruments CC-40

The Texas Instruments Compact Computer 40 (CC-40) bridges the gap between the pocket and notebook categories. It has one of the best versions of Basic ever offered by TI and provides exceptional accuracy, albeit rather slowly.

The CC-40 is powered by four AA alkaline batteries which will last for 200 hours of use, considerably longer than most other notebook machines. If you prefer, an AC adapter is also available.

Although the keyboard is arranged in the standard QWERTY layout, it is only two-thirds the size of a standard keyboard and sports calculator-style keys. Thus, touch typing is not possible, and even experienced typists will find a two-finger approach more reliable. For data entry, a numeric keypad is provided to the right of the main keyboard.

The CC-40 uses a single line, 31-character display capable of reproducing upper- and lowercase text and a variety of graphics symbols. The display scrolls horizontally on a maximum 80-character line. The LCD screen also displays several special status indicators above and below the main text line, so it is more versatile than it might appear.

As it comes out of the box, the CC-40 does not interface to anything directly—not even a cassette recorder. However, an eight-pin connector attaches to a hex-bus peripheral module. This unit provides an interface to three peripherals, an RS-232 interface, a printer/plotter, and a wafertape drive. On the top of the CC-40 is a cartridge port that can accept ROM cartridge software or a memory expansion cartridge.

TI has announced a wide variety of software packages on both wafertape and ROM cartridges. The packages tend to be adaptations of programmable calculator software and will have greatest appeal for engineers and financial analysts.

The most likely market for the TI CC-40 is probably as a competitor to pocket computers and as an upgrade from programmable calculators. In this market with its low price. tag (frequently discounted to under $200), it is a formidable competitor.

Casio FP-200

Casio is a very successful maker of calculators, watches, and electronic musical keyboards. However, their previous forays into computers have ended in failure, at least in the U.S. With the FP-200, they have taken a different approach and may be able to carve out their own niche in the market.

The FP-200 is primarily a spreadsheet machine and runs a built-in software package called CETL (Casio Easy Table Language). It is a *VisiCalc*-like language, and anyone familiar with another spreadsheet will be able to use it immedi-

Texas Instruments CC-40

Casio FP-200

WorkSlate

Canon X-07

ately. Also built-in is Casio Basic, a Microsoft-like implementation with rather leisurely performance.

The FP-200 is built around a CMOS version of the Z80 and has 32K of ROM and 8K of RAM, expandable to 32K. External mass storage requires either a cassette tape recorder (300 baud) or a 70K single-sided, single-density floppy disk drive. Output ports are also provided for a parallel printer and RS-232 serial device such as a modem.

The FP-200 has an 8-line x 20 character display. For graphics, 64 x 160 pixels can be individually addressed.

The keyboard is full-size, but uses calculator-style short-throw keys. It has 57 alphanumeric keys, four special keys, five function keys (two meanings each), and four cursor keys (arranged in a straight line). An optional, external numeric keypad is available.

The FP-200 would be a good choice if you are looking for a machine primarily for spreadsheet calculations and some Basic programming. The keyboard renders it unsuitable for word processing, even if there were software available which there is not. Several utility software packages (sort/merge, statistics, graphics) are promised and would further enhance the utility of the machine.

The suggested retail price is $499, but we have yet to see a machine in any retail outlets.

WorkSlate

Like the Casio FP-200, the WorkSlate from Convergent Technologies, is primarily a spreadsheet machine. Indeed, all the software packages on the WorkSlate are various adaptations of the basic spreadsheet program.

In a departure from all of the other notebook computers, the WorkSlate uses a CMOS version of the 6800 mpu. Included with the basic system is 16K of RAM memory. Currently, this is not expandable, but a company spokesman tells us that a 16K upgrade (32K total) will eventually be available. That is necessary, because the 16K gets filled up with just a 30 x 24 cell spreadsheet; in this state, it takes over two minutes just to perform a simple operation. Theoretically, a spreadsheet can expand to 128 rows and columns, but if you have 128 in either dimension, 5 is the maximum in the other.

The display on the WorkSlate consists of 16 lines by 42 characters. Some lines are devoted to status indicators, headings, and formulae; as a result, about 11 by 5 cells of a spreadsheet are visible at a time.

The keyboard follows the QWERTY layout, but it is about six percent smaller than a standard keyboard. This is not bothersome, but the circular keys with less than full travel might be. Clearly, it is not designed for text entry, and the majority of the numeric entries will be made from the numeric keypad to the right of the main keyboard. Indeed, there is no row of numeric keys over the alphabetic portion of the keyboard. A nice touch is the large diamond shaped cursor control key between the alpha and numeric keys.

To the right of the display is a microcassette recorder. This is a dual-function unit which can record either data or audio. The data transfer rate is 2400 baud, one of the fastest available on a small computer. For audio recording, WorkSlate has a built-in speaker, microphone, and jacks for an external earphone and microphone.

Also built in are a direct-connect, 300-baud modem and communications software. A matching printer, capable of 40- or 80-column compressed print widths is available for $250. Alternatively, an adapter is available for standard parallel or RS-232 serial printers or other devices.

We mentioned software packages other than the basic spreadsheet. These are Memo Pad, Phone List, and Calendar. Actually, the only difference between these and blank spreadsheets are some headings, graphics, and modifications of column width. In other words, a Memo Pad is simply a spreadsheet whose A column is 128 characters wide. Two self-instruction tutorial tapes are also available.

As the reviewer in our sister publication, *Computers & Electronics* concluded, "The features of WorkSlate are geared to the business person who wants spreadsheet capabilities in a portable package without all the fuss of learning about computers. For those who want such a business tool, WorkSlate has hit its design mark."

Canon X-07

The Canon X-07 is an exceptionally compact computer, weighing just over one pound. It has a four-line display, sensible cursor movement keys, and a novel optical communications capability.

The X-07 is primarily intended for computing in Basic, and no other software is currently built-in or offered. The default mode for calculations is double precision; as a result, the machine is quite accurate, but was the slowest of the 14 tested.

The X-07 uses a CMOS version of the Z80 mpu and has 20K of ROM and 8K of RAM (expandable to 24K). An external cassette recorder provides mass storage. The X-07 is powered by four AA akaline batteries with a rated life of "up to 2000 hours of use." Unbelievable!

Small memory cards about the size of two stacked credit cards are available with both ROM for applications software packages and RAM for removable user memory. Each card has a lithium watch battery which provides power for over a year.

The keyboard is laid out in QWERTY fashion, but it is 20% smaller than a standard keyboard and has calculator-

Hewlett Packard HP-75C

Toshiba T100

style keys. The spacebar is half the normal size, and there is only one SHIFT key, located below the Z. Thus, although it has a good feel, it is not suitable for rapid typing. However, for program editing, the cursor control keys are ideal, being oversize and laid out in a logical pattern.

The display has four lines of 20 characters each. Also, individual pixels can be addressed on the 32 x 120 screen. The X-07 also has a speaker capable of reproducing notes over four octaves.

Six peripherals are available for the X-07. A color plotter both prints and plots on 4½" wide paper in four colors. A compact thermal printer uses narrow (2¼") thermal paper. Three interface modules are available: one for RS-232 devices, one for a monitor or TV set, and another for interfacing to parallel devices such as Centronics-type printers, sensors, and synthesizers.

The last peripheral is an odd one; it is an infra-red optical coupler that permits two X-07 computers to communicate with each other. It makes a novel demonstration at trade shows, but we're not sure of what use it would be to the average user.

The Canon X-07 is a capable Basic-speaking computer in a compact package with a nice printer/plotter available. Support may be another issue. All of our testing of the X-07 was done in France and Japan where the Canon people were much more helpful than in the U.S. Buyers should hope that Canon will be more friendly to them than they are to the press.

Hewlett Packard HP-75C

The HP-75 is a very compact computer with unexpected speed and the highest accuracy of any computer we have ever tested—portable or desktop. It has a one-line display, but a monitor, as well as several other peripherals, are available.

At first glance, the keyboard appears to be much smaller than a standard one, but in fact is only 5% reduced. However, it uses calculator-style, short-travel keys and has a few keys in odd places; thus it is not suitable for rapid typing. A block of ten alpha keys can be specified as a numeric keypad for data entry.

Like earlier HP calculators, the HP-75 uses magnetic cards for program and data storage. Each card is 10" long and stores up to 1.4K. The cards must be pulled by hand through a slot on the lower right; it takes a bit of practice to get the hang of it, but the computer tells you if you have pulled it too fast or too slow.

The HP-75 uses a custom HP CMOS processor. It comes with 8K of RAM built in. Three slots on the front accommodate additional 8K RAM cartridges or ROM software cartridges. The machine can hold several programs which are accessed like a mini timesharing system. The HP-75 even has an Appointment Mode which can trigger one of nine different alarm sounds, a Basic program, or other action.

As expected, the HP-75 uses a dialect of HP Basic which is quite different from the DEC Basic, on which Microsoft Basic was modeled. Thus, string handling, functions, and PRINT USING (DISPLAY USING on the HP-75) are quite different from what much of the world regards as standard. Nevertheless, HP Basic is quite satisfactory and has several unexpected features such as powerful TRACE utility, recursive calls, and the ability to issue operating system commands from within a Basic program.

As might be expected, HP is converting programmable calculator application packages to the HP-75. Several engineering and financial packages have been released, and more are on the way. Even *VisiCalc* is available for the HP-75; with the single 32-character display line, it is less than satisfactory, but with the optional monitor hookup, it is fine. A word processing module has been announced, but we can't imagine doing serious text entry on the HP-75 keyboard.

The HP-75 uses an interface loop structure with each peripheral device daisychaincd in a continuous loop. Available peripherals include a digital cassette drive (fast and reliable), 24-column thermal printer, video interface (16 lines x 32 characters), plotter, 80-column dot matrix printer, and several laboratory device controllers.

The HP-75 is a fine computer system, particularly for a laboratory user or someone stepping up from a programmable calculator. It is well-engineered, has a good Basic (albeit a bit unusual), and a nice array of immediately available peripherals.

Toshiba T100

Unlike the other 13 machines in this roundup, the Toshiba T100 was not designed primarily as a notebook portable. Rather, it is the keyboard/system unit of an excellent desktop system to which attaches an eight-line LCD display and acoustic coupler to make it portable. It still needs AC power. Hence, it is primarily targeted at people who want to do all their computing on the same machine, but occasionally require computing capability away from the home office.

The T100 uses a Z80A running at 4 MHz; it was the fastest 8-bit machine we have tested to date. It has 32K of ROM and 64K of RAM built in. Two slots at the upper right accept ROM and RAM packs which are accessed from CP/M as disk drive E.

The keyboard is full-size with 89 full-stroke keys divided into a standard keyboard, numeric keypad, six special keys,

Teleram 3000


Radio Shack Model 100

and eight function keys (two user-programmable meanings each).

Three display connectors are provided, one each for a monochrome and RGB color monitor and one for the 8-line x 40-character LCD display. Individual pixels (64 x 240) can be addressed on the LCD display, a far cry from the spectacular 240 x 600 pixel resolution available on the color monitor.

For communications capability in either portable or stationary mode, the T100 uses the Lexicon LEX-12 modem, a 300-baud combination direct-connect/acoustic modem.

As mentioned, the T100 comes with CP/M 2.2 and Microsoft Basic in ROM. A disk, furnished with the portability package, scales the various menus to the LCD display and provides an assortment of utility packages. You can load any of these programs onto a RAM cartridge.

The T100 comes with an impressive array of bundled software including *Word Right, Magic Worksheet, MatheMagic, GraphMagic, Analyst, Q-Sort,* and *NAD.* Not all of these are suitable for use in the portable mode since several require disk operations.

The T100 has been on sale in Japan for over a year and is a proven performer. Toshiba offers a good array of peripherals including disk drives, monitors, printers, and a hardside carrying case for the cpu unit, display, and modem. The T100 should appeal to the user who wants, or can only afford, one computer; who needs an outstanding desktop machine; but who also needs limited away-from-home capability.

Teleram 3000

The Teleram 3000 was the first notebook portable introduced. It was targeted at large corporate users, and has found good acceptance in that market.

The 3000 is one of the largest notebook machines, weighing in at nine pounds. However, in this package, it has a full-size, full-stroke, completely standard keyboard; four-line by 80-character display; 128K of internal bubble memory (expandable to 256K); and CP/M operating system.

The 3000 has 64K of RAM; the one or two 128K bubble memory cartridges are accessed from CP/M like a floppy disk in drive A. The Teleram uses a CMOS version of the Z80; performance is similar to the four other machines using this configuration.

In addition to the standard alphanumeric keyboard, the 3000 has a numeric keypad to the right, eight programmable function keys with two meanings each, and several special keys.

Built into the 3000 are MBasic, several CP/M utilities, and a communications program called teleTalk. Microsoft Basic functions are as expected, although the graphics commands are not implemented; apparently the Teleram people couldn't imagine doing graphics on a 1.1" x 8.2" LCD screen—can't say as we blame them.

The teleTalk package is an especially rich communications package providing auto-dial, auto-answer, data capture, dump, and file transfer. Telephone numbers, passwords, commands, and log-on procedures may all be stored in command files. We were very impressed with the file transfer capabilities which allow all kinds of files, even CP/M .COM and text files, to be sent, received, and processed.

The Teleram 3000 has exceptional communications capabilities, a good Basic, and a standard operating system (CP/M) which opens the door to a large library of programs. Thus, the machine should have greatest appeal to the executive on the move, and the large company target audience of Teleram makes good sense.

Radio Shack Model 100

The story of the Radio Shack TRS-80 Model 100 is actually not the story of a TRS-80 at all. The machine was conceived by a small Japanese company, Kyoto Ceramics and first sold in Japan by NEC. Radio Shack designers worked with Kyocera to incorporate certain changes and additional features in the computer before introducing it in the American market. These changes were apparently "right," as few other computers have enjoyed such runaway sales success as the Model 100.

The Model 100 is truly notebook size (8.5" x 12" x 2.2") and weighs just under four pounds. It incorporates a full-size, full-stroke keyboard, with four special keys, eight function keys, and four cursor keys (in an unfriendly straight line).

The display is the largest on any notebook computer, 2" x 7.5", and displays eight lines of 40 characters each. The character size is large and legible. Graphics within a 64 x 240 pixel matrix are also possible. A built-in speaker plays notes over a five-octave range.

The Model 100 uses a CMOS version of the Z80 running at 2.5 MHz. Since default mode in Basic is double precision, the machine was very slow in running our benchmark; on the other hand, it scored high in the accuracy department. It has only 8K of RAM built in, but a 24K version is available. Both can be further expanded to 32K. An external cassette recorder provides mass storage.

The computer provides an impressive array of I/O ports. On the back are connectors for Centronics parallel printer, RS-232 serial device, cassette recorder, bar code reader, and modular telephone jack.

The Model 100 has a built-in direct-connect modem which can plug into any telephone jack. Coupled with the communications software package, it pro-

NEC PC-8201

Epson HX-20

vides many of the features of a so-called "smart" modem—auto-dial, log-on, download, and upload—although it does not have wake up and auto-answer.

The Model 100 has five programs built in. Microsoft Basic is missing a few commands and does not have on-screen editing (except by means of the text editor, a cumbersome process). The text editor is an adequate package. It is always in insert mode, and has cut, paste, search, and other rudimentary features. It does not have an output formatter, but several are available from third party vendors.

The communications package was mentioned above. The last two packages, schedule organizer and name/address organizer, are simply special versions of the text editor with certain commands locked out. We have not found them particularly useful.

Many software packages have been introduced already by third party vendors, and much more is on the way. The availability of software coupled with the integrated packages built into the machine make the Model 100 an attractive choice for a wide variety of users. Poor Basic program editing and lack of an output formatter are small drawbacks against the many enticing capabilities of the computer coupled with an attractive price.

NEC PC-8201

We have frequently called the NEC PC-8201 the twin of the Radio Shack Model 100. Strictly speaking, this is not true. The 8201 was born six months or so earlier in Japan and is a somewhat different version of the Kyoto Ceramics original.

The 8201 is slightly larger than the Model 100 as a result of providing a slot in the left side for an expansion memory cartridge. The 8201 can have from 16K to 64K of RAM in the basic machine. The 32K plug-in memory cartridges function as a switchable bank of main memory, rather than a disk drive as on some other machines.

The 8201 has a full-size, full-stroke keyboard with two special keys, five function keys (two programmable meanings for each), and cursor control keys laid out in a logical diamond pattern. Graphics characters are not built into the 8201 as they are on the Model 100; instead any desired graphics characters can be entered by the user with a short included utility program.

The display is 2" x 7.5" and displays eight lines of 40 characters each. The version of Microsoft Basic on the NEC has the LOCATE command not found on the Model 100, so characters as well as individual pixels (64 x 240) can be addressed. The Basic on the NEC also has on-screen editing and RENUM, both lacking in the Model 100 implementation.

In addition to Basic, the 8201 has a text editor (but no output formatter) and telecommunications software package (but no built-in modem). The 8201 also comes with a cassette tape of 16 utility and demonstration programs—a desirable extra.

The 8201 has output connectors for Centronics parallel printer, RS-232 serial device, bar code reader, modem, and cassette recorder. The literature promises a floppy disk interface, but we have been unable to get any information on it.

For an attractive price, the NEC PC-8201 offers a great deal of capability—Basic computations, word processing, and communications, with RAM cartridges for external storage. The machine should have appeal for a wide cross section of people needing a true portable computer.

Epson HX-20

The Epson HX-20 was the first true notebook size computer introduced. Unfortunately, lack of availability prevented it from being a runaway success when it was introduced in late 1982. Now that it is widely available, it no longer has the market to itself and will have to carve out a smaller niche among a tough field of competitors.

Several reviewers have looked at the small screen size of the HX-20 and concluded that it is not competitive with the later entries sporting screens four to eight times larger. We think that is unfair, as the HX-20 still offers a wide array of features, some of them unique in a machine this size.

The Epson is slightly thinner than the Model 100, but weighs the same 3.8 pounds. It has a full-stroke, standard size keyboard with an excellent feel. Along with the 54-key QWERTY keyboard, it has seven special keys and five function keys, each with a dual meaning. The main disadvantage is that there are only two cursor control keys; the other two directions are gotten by pressing shift with one of the two keys. Although Basic has on-screen editing, using just two keys is a pain.

The LCD screen displays four lines of 20 characters each. The display is actually a window onto a much larger virtual screen; the size can be specified by the user. Hence, it is possible to scroll in both directions. Pixel and character addressing are possible within the 32 x 120 pixel dimensions of the screen. A small speaker can produce tones over a four-octave range.

The HX-20 uses a CMOS version of the Z80 mpu. It has 32K of ROM and 16K of RAM, expandable to 32K with an external module. Mass storage is provided in the form of a built-in microcassette recorder. We found this to be fast and reliable. An external cassette can also be used.

The HX-20 provides I/O connectors for RS-232 serial devices, bar code reader, cassette recorder, and a 38,400-

RoadRunner

Sharp PC-5000

baud serial link to other devices via an interface module which has yet to be released.

On the top left of the case is a built-in printer. It uses plain paper rolls $2\frac{1}{4}$" wide and prints in black or purple. It is this printer that makes the small screen size tolerable as programs or text can be printed out in rough form for correction and then printed later on a full-size printer or transmitted to another machine.

The NiCad rechargeable battery on the HX-20 provides 50 hours of use, considerably more than any of the other notebook portables.

Built into ROM is a rudimentary monitor, Microsoft Basic, and *SkiWriter*, a word processing package. Basic is a complete implementation with no obvious omissions. Up to four Basic programs can be stored simultaneously in the machine. Many more, of course, can be stored on tape.

SkiWriter is an adequate, if not extensive, word processing package. It can operate in either insert or overstrike mode and has block copy and delete. It will search for a string, but will not search and replace automatically. Print formatting is barely adequate, as it requires that you put page breaks into the text rather than producing them automatically.

Epson has announced a wide array of plug-in ROM software packages, but we have not seen them at the retail level yet. The communications and spreadsheet packages should enhance the appeal of the machine considerably.

The HX-20 with built-in printer and optional microcassette offers a great deal of computing power at a moderate price. The long battery life between charges is a nice plus. The machine should appeal to people needing a full-feature Basic and occasional word processing. Additional software packages should help it carve out a niche with specific types of users.

RoadRunner

The MicroOffice RoadRunner is one of the latest entries in the notebook computer sweepstakes. Currently, it is being marketed primarily to OEMs and large-volume end users, but it may be available by mail order and in a limited number of retail computer stores.

The RoadRunner is equal in size to a large binder and weighs in at five pounds. When it is opened, it comes to life with a small beep and initial dialog on the display.

The LCD screen measures 1.3" x 9.3" and displays eight lines of 80 characters, about the size of a dot matrix printer. Graphics can be displayed on the 64 x 480 pixel screen. The display tilts, but unfortunately does not have a contrast adjustment.

The keyboard is a full-size, full-stroke unit with sculpted concave keys. Although close to a standard layout, it has several keys in unexpected locations. Most users will probably adjust in a week or two. The cursor keys are laid out in a logical pattern, an oversight on too many notebook portables. Six special keys and eight dual-meaning function keys are found in the top row above the standard QWERTY keyboard.

The RoadRunner uses a CMOS version of the Z80 mpu and has 16K of ROM and 48K of RAM. Four memory cartridge slots are found over the keyboard for extra RAM memory and ROM software cartridges. These are addressed from the CP/M-compatible operating system as devices A through D.

I/O connectors are provided for RS-232 serial devices, a modem module, and the main bus of the computer. The modem is a 300-baud direct-connect unit with auto-dial, auto-answer, and a wake-up mode of operation. In addition, the RoadRunner has a built-in terminal mode which emulates a DEC VT100 terminal.

Also built in is a schedule organizer, name/address organizer, and, of course, the CP/M operating system. Available on cartridge is a full-feature word processing package with features such as character, word, and line delete, and global search-and-replace.

Cartridges are also available with Microsoft Basic (with everything but on-screen editing) and Sorcim *SuperCalc*. More packages are promised in the future.

With its excellent communications, word processing, Basic, and spreadsheet software on a machine with full keyboard, large display, and plug-in memory cartridges, the RoadRunner is sure to find a market with executives, sales people, and writers who need a portable office on the road.

Sharp PC-5000

The Sharp PC-5000 is one of the largest of the notebook computers, but it is packed with features and capability. It has a large screen (eight lines by 80 characters), full keyboard, 16-bit processor, 128K memory, and much more.

The PC-5000 uses a 16-bit 8088 mpu, the same as in the IBM PC. MS-DOS

Xerox 1810

and Microsoft GW (Gee Whiz) Basic reside in 64K of ROM, and 128K of RAM is available for user memory, expandable to 256K. Mass storage is in the form of a 128K bubble memory cartridge or, if you prefer, an external cassette recorder. In a non-portable mode, the PC-5000 also supports a double-density, double-sided floppy disk drive.

With the 16-bit processor, the PC-5000 is fast—close to the fastest computer we have ever tested, portable or not. A second control mpu controls the I/O functions, further contributing to the throughput.

The LCD screen measures 1.3" x 9.3" and displays eight lines of 80 characters or graphics in an 80 x 640 pixel field. Characters are about the same size as dot matrix printer output—small, but readable.

The keyboard has 57 full-stroke keys, three special keys, eight dual-meaning function keys, and four cursor control keys (laid out in a straight line, unfortunately). The keys are concave sculpted, and have a good feel, except for a slight "give" in the center of the keyboard.

Connectors are provided for a cassette recorder, external bus, RS-232 serial device, and modem. The modem is an option with the PC-5000 and fits into the lid of the case. It is a 300-baud, direct-connect unit with an auto-dial, redial, and conference phone capability. It is supported by the *SuperComm* software package by Sorcim.

Another optional extra which fits in the basic unit is an 80-column thermal printer which can print on plain paper as well as on thermal paper. It is a 30 cps unit that produces excellent type (and dot graphics) in a variety of formats.

Software is available on bubble memory cartridges or on disk. In addition to the built-in MS-DOS and Microsoft GW Basic, external software includes *SuperWriter*, an excellent menu-driven word processing package by Sorcim; *SuperCalc-2*, a spreadsheet package; and *SuperComm*. Many other software houses are said to be working to fit their packages on the PC-5000.

The PC-5000 is an outstanding, full-featured computer with a variety of excellent software backed up by some innovative serving arrangements. As such, it should find enthusiastic acceptance by a wide cross section of business people and other people on the move who need full-function computing power.

Xerox 1810

The Xerox 1810 is a notebook portable designed by Sunrise Systems. It is sold (by them) only to OEMs. Currently, the only OEM actively marketing the system is Xerox; from them it is known as the Xerox 1800. It is likely to be available from other vendors in slightly different configurations in the near future.

Like the Toshiba T100, the 1810 is the keyboard, cpu, and system unit of a desktop computer, although the LCD display is built in. The machine can be customized to the particular needs of an individual vendor, thus some vendors may offer it as a desktop unit with portable cpu, and others may offer it just as a notebook portable. Xerox offers both the 1810 keyboard unit plus an optional 1850 "Flat Pack" processor.

The processor is a CMOS version of the Z80 with 32K of ROM and 16K of RAM, expandable to 64K. In the desktop configuration, the unit contains a dual 16-bit 8088 and an 8-bit Z80 plus additional memory. Mass storage in the portable mode of operation is provided by a built-in microcassette recorder to the right of the keyboard. This stores 512K of digital information and also doubles as an audio recorder.

The keyboard has 63 full-stroke keys in a more or less standard layout. The


Gavilan

cursor control keys are arranged logically, although the up cursor key is where you might expect to find the question mark on the bottom row. A row of special and programmable function keys is over the regular keyboard.

The LCD display is available in two versions: six lines by 40 characters or three lines by 80 characters. For some inexplicable reason, Xerox chose the latter configuration for the 1810. Individual pixels are addressable, although hardly useful in the 3 x 80 layout (24 x 480 pixels). The unit also provides a signal for a monitor.

A direct-connect 300-baud modem, supported by a dumb terminal software package, is built in. An interesting telephone program lets the computer be used as an answering machine, and provides auto-dial, redial last number, hold, and speakerphone capabilities.

I/O connectors are provided for RS-232 serial devices, Centronics parallel printer, telephone, and RGB or monochrome monitor.

A vast array of software is available and under development. Built in is CP/M, a calendar/scheduler, and four-function calculator. Software on plug-in ROM cartridges includes Microsoft Basic, typewriter/note taker (a word processing package), a terminal program, and the telephone program mentioned above.

The Xerox 1810 is a state-of-the-art machine with a wide array of software and easy access to much more as a result of the CP/M operating system and communications capability. We will be anxious to see how Xerox (and other vendors) target it.

Gavilan

When the Gavilan was first shown at NCC in June 1983, many people wondered out loud whether it could really be made. It was just too much state-of-the-art stuff in one package for people to swallow. Now, that it is nearing readiness for the market and now that competitors are leaping in from all directions, the Gavilan looks more real than it did six months ago. But it still has an air of unreality.

The Gavilan is remarkably compact (11.4" square x 2.7" high) and weighs just nine pounds. Its optional, snap-on printer adds four inches and four pounds to the package. It has a 16-bit mpu, 64K of RAM, built-in 3" floppy disk, eight-line display, full keyboard, and a unique touch panel in which your finger becomes sort of an electronic mouse.

The Gavilan uses a 16-bit 8088 mpu, 48K of ROM, and 64K of RAM, expandable with up to four 32K plug-in capsules of blank memory or applications software packages. Also built in is a 3", 320K Hitachi floppy disk drive. Traffic cop chips turn off the power to the disk drive or mpu whenever it is not needed; thus the NiCad batteries provide eight hours of operation. Eighty percent of the charge can be restored in one hour on the charger.

The keyboard is a standard-size, full-stroke unit with 13 special keys to the right. Several keys are in "unusual" locations, especially the enter key next to the spacebar, but it shouldn't take too long to get used to it.

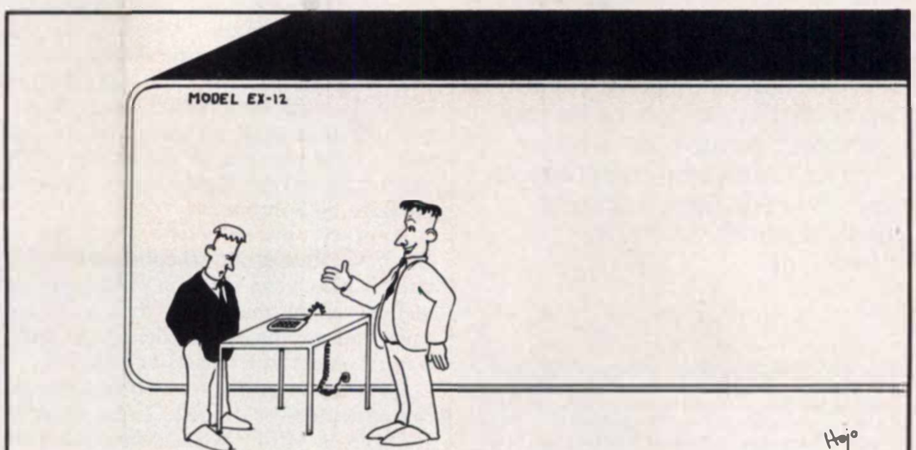
The most unusual feature of the Gavilan is the touch pad below the display. This lets you manipulate objects on the screen by pointing at them. A quick movement of your finger moves the cursor a long way while a slow movement gives you fine control. Like Apple's Lisa system, pictorial representations of objects such as file drawers, file folders, documents, and a trash basket are shown on the screen.

Although the screen is capable of displaying eight lines of 80 characters, in most cases, part of the screen will be devoted to menus in "windows" appropriate to the software package currently in use. This loss of a portion of the screen makes one wish mightily for a screen three times as large. At home base your prayers are answered, since the Gavilan provides standard video output to a monochrome monitor.

The Gavilan has a built-in 300-baud direct-connect modem supported by a comprehensive software package. It also has interfaces for the optional printer, a second disk drive, RS-232 serial device, and video output.

Besides MS-DOS, MBasic, and the communications software, Gavilan makes available an Office Pack of four applications, Sorcim *SuperCalc* and *SuperWriter*, and *PFS File* and *Report*.

All in all, the Gavilan offers as much or more than most desktop systems. It is a full-function computer with few trade-offs. All this comes at a price, but for the traveling professional this is a machine that is easy to learn and exceptionally user-friendly, and one that will spark envy in the eyes of all who see it. ■



"... and once you're ready to use this computer with software all you'll need is the expansion module."

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